

STAT 400 Homework 1: Calculus Review

1) Consider the following:

$$S = \{L, A, S, E, R\}$$

a) List out all possible subsets of S .

$$\begin{array}{ccccccccc} & & & & \emptyset & & & & \\ \{L\} & & \{A\} & & \{S\} & & \{E\} & & \{R\} \\ \{L, A\} & \{L, S\} & \{L, E\} & \{L, R\} & \{A, S\} & \{A, E\} & \{A, R\} & \{S, E\} & \{S, R\} & \{E, R\} \\ \{L, A, S\} & & \{L, A, E\} & & \{L, A, R\} & & \{L, E, S\} & & \{L, R, S\} \\ \{E, A, S\} & & \{R, A, S\} & & \{R, E, S\} & & \{R, E, L\} & & \{R, E, A\} \\ \{L, A, S, E\} & & \{L, A, S, R\} & & \{L, A, R, E\} & & \{L, R, S, E\} & & \{R, A, S, E\} \\ & & & & \{L, A, S, E, R\} & & & & \end{array}$$

b) For $n = 0, 1, 2, 3, 4, 5$, count the number of subsets of size n .

- There is **1** subset of size 0.
- There are **5** subsets of size 1.
- There are **10** subsets of size 2.
- There are **10** subsets of size 3.
- There are **5** subsets of size 4.
- There is **1** subset of size 5.

2) Recall that the function $f(x) = \frac{1}{1-x}$ has the power series expansion

$$\sum_{n=0}^{\infty} x^n$$

For any $p \in (0, 1)$, calculate:

a)

$$\sum_{n=1}^{\infty} p(1-p)^{n-1}$$

$$a = p(1-p)^0 = p$$

$$r = 1 - p$$

$$\begin{aligned} \sum_{n=1}^{\infty} p(1-p)^{n-1} &= \frac{a}{1-r} \\ &= \frac{p}{1-(1-p)} \\ &= \frac{p}{p} \\ &= 1 \end{aligned}$$

b)

$$\sum_{n=1}^{\infty} np(1-p)^{n-1}$$

$$a = p(1-p)^0 = p$$

$$r = 1-p$$

$$\begin{aligned} \sum_{n=1}^{\infty} np(1-p)^{n-1} &= \frac{d}{dr} \left(\sum_{n=1}^{\infty} pr^n \right) \\ &= \frac{d}{dr} \left(\frac{a}{1-r} \right) \\ &= \frac{a}{(1-r)^2} \\ &= \frac{p}{(1-(1-p))^2} \\ &= \frac{1}{p} \end{aligned}$$

c) Use the above information to calculate

$$\sum_{n=1}^{\infty} n \left(\frac{1}{2} \right)^n$$

$$a = \left(\frac{1}{2} \right)^1 = \frac{1}{2}$$

$$r = \frac{1}{2}$$

$$\begin{aligned} \sum_{n=1}^{\infty} n \left(\frac{1}{2} \right)^n &= \frac{d}{dr} \left(\sum_{n=1}^{\infty} r^{n+1} \right) \\ &= \frac{d}{dr} \left(\sum_{n=1}^{\infty} r \cdot r^n \right) \\ &= \frac{d}{dr} \left(a \frac{r}{1-r} \right) \\ &= a \left(\frac{(1-r) + r}{(1-r)^2} \right) \\ &= a \left(\frac{1}{(1-r)^2} \right) \\ &= \frac{a}{(1-r)^2} \\ &= \frac{1/2}{(1-1/2)^2} = \mathbf{2} \end{aligned}$$

3) Let $f(x) = e^{-3x}$, calculate the following:

a)

$$\begin{aligned}\int_0^\infty f(x) dx &= \lim_{b \rightarrow \infty} \int_0^b e^{-3x} dx \\ &= \lim_{b \rightarrow \infty} \left(-\frac{1}{3} e^{-3x} \right) \Big|_0^b \\ &= \lim_{b \rightarrow \infty} \left(-\frac{1}{3} e^{-3b} + \frac{1}{3} e^0 \right) \\ &= \lim_{b \rightarrow \infty} \left(-\frac{1}{3} e^{-3b} + \frac{1}{3} \right) \\ &= \frac{1}{3}\end{aligned}$$

b)

$$\int_0^\infty x f(x) dx = \lim_{b \rightarrow \infty} \int_0^b x e^{-3x} dx$$

$$u = x \quad dv = e^{-3x} dx$$

$$du = dx \quad v = -\frac{1}{3} e^{-3x}$$

$$\begin{aligned}\lim_{b \rightarrow \infty} \int_0^b x e^{-3x} dx &= \lim_{b \rightarrow \infty} \left(\left(-\frac{1}{3} x e^{-3x} \right) \Big|_0^b - \int_0^b -\frac{1}{3} e^{-3x} dx \right) \\ &= \lim_{b \rightarrow \infty} \left(-\frac{1}{3} b e^{-3b} - \int_0^b -\frac{1}{3} e^{-3x} dx \right) \\ &= \lim_{b \rightarrow \infty} \left(-\frac{1}{3} b e^{-3b} - \left(\frac{1}{9} e^{-3x} \right) \Big|_0^b \right) \\ &= \lim_{b \rightarrow \infty} \left(-\frac{1}{3} b e^{-3b} - \left(\frac{1}{9} e^{-3b} - \frac{1}{9} \right) \right) \\ &= \lim_{b \rightarrow \infty} \left(-\frac{1}{3} b e^{-3b} - \frac{1}{9} e^{-3b} + \frac{1}{9} \right) \\ &= \frac{1}{9}\end{aligned}$$

c)

$$\int_0^{\infty} x^2 f(x) dx = \lim_{b \rightarrow \infty} \int_0^b x^2 e^{-3x} dx$$

u	dv
x^2	e^{-3x}
$2x$	$-\frac{1}{3}e^{-3x}$
2	$\frac{1}{9}e^{-3x}$
0	$-\frac{1}{27}e^{-3x}$

$$\begin{aligned} \lim_{b \rightarrow \infty} \int_0^b x^2 e^{-3x} dx &= \lim_{b \rightarrow \infty} \left(-\frac{1}{3}x^2 e^{-3x} - \frac{2}{9}x e^{-3x} - \frac{2}{27}e^{-3x} \right) \Big|_0^b \\ &= \lim_{b \rightarrow \infty} \left(-\frac{1}{3}b^2 e^{-3b} - \frac{2}{9}b e^{-3b} - \frac{2}{27}e^{-3b} - \left(-\frac{2}{27}e^0 \right) \right) \\ &= \frac{2}{27} \end{aligned}$$

4) Consider the function $f(x, y) = e^{-(2x+5y)}$. Calculate

$$\iint f(x, y) dy dx$$

where the region of integration is:

a) The rectangle $[0, 3] \times [1, 3]$.

$$\begin{aligned} \int_0^3 \int_1^3 e^{-(2x+5y)} dy dx &= \int_0^3 \left(-\frac{1}{5}e^{-2x-5y} \right) \Big|_1^3 dx \\ &= \int_0^3 \left(-\frac{1}{5}e^{-2x-15} + \frac{1}{5}e^{-2x-5} \right) dx \\ &= \left(\frac{1}{10}e^{-2x-15} - \frac{1}{10}e^{-2x-5} \right) \Big|_0^3 \\ &= \frac{1}{10}e^{-21} - \frac{1}{10}e^{-11} - \left(\frac{1}{10}e^{-15} - \frac{1}{10}e^{-5} \right) \\ &= \frac{1}{10}e^{-21} - \frac{1}{10}e^{-15} - \frac{1}{10}e^{-11} + \frac{1}{10}e^{-5} \end{aligned}$$

b) The rectangle $[0, 3] \times [0, 3]$.

$$\begin{aligned}
 \int_0^3 \int_0^3 e^{-(2x+5y)} dy dx &= \int_0^3 \left(-\frac{1}{5} e^{-2x-5y} \right) \Big|_0^3 dx \\
 &= \int_0^3 \left(-\frac{1}{5} e^{-2x-15} + \frac{1}{5} e^{-2x} \right) dx \\
 &= \left(\frac{1}{10} e^{-2x-15} - \frac{1}{10} e^{-2x} \right) \Big|_0^3 \\
 &= \frac{1}{10} e^{-21} - \frac{1}{10} e^{-6} - \left(\frac{1}{10} e^{-15} - \frac{1}{10} e^0 \right) \\
 &= \frac{1}{10} e^{-21} - \frac{1}{10} e^{-15} - \frac{1}{10} e^{-6} + \frac{1}{10}
 \end{aligned}$$

c) The region bounded by the line $y = 2x$ and the x -axis, for $x \in [0, 3]$.

$$\begin{aligned}
 \int_0^3 \int_0^{2x} e^{-(2x+5y)} dy dx &= \int_0^3 \left(-\frac{1}{5} e^{-2x-5y} \right) \Big|_0^{2x} dx \\
 &= \int_0^3 \left(-\frac{1}{5} e^{-12x} + \frac{1}{5} e^{-2x} \right) dx \\
 &= \left(\frac{1}{60} e^{-12x} - \frac{1}{10} e^{-2x} \right) \Big|_0^3 \\
 &= \frac{1}{60} e^{-36} - \frac{1}{10} e^{-6} - \left(\frac{1}{60} - \frac{1}{10} \right) \\
 &= \frac{1}{60} e^{-36} - \frac{1}{10} e^{-6} + \frac{1}{12}
 \end{aligned}$$

d) The region bounded by the line $y = 2x$ and the x -axis, for $x \in [0, \infty]$.

$$\begin{aligned}
 \int_0^\infty \int_0^{2x} e^{-(2x+5y)} dy dx &= \int_0^\infty \left(-\frac{1}{5} e^{-2x-5y} \right) \Big|_0^{2x} dx \\
 &= \int_0^\infty \left(-\frac{1}{5} e^{-12x} + \frac{1}{5} e^{-2x} \right) dx \\
 &= \lim_{b \rightarrow \infty} \int_0^b \left(-\frac{1}{5} e^{-12x} + \frac{1}{5} e^{-2x} \right) dx \\
 &= \lim_{b \rightarrow \infty} \left(\frac{1}{60} e^{-12x} - \frac{1}{10} e^{-2x} \right) \Big|_0^b \\
 &= \lim_{b \rightarrow \infty} \left(\frac{1}{60} e^{-12b} - \frac{1}{10} e^{-2b} - \left(\frac{1}{60} - \frac{1}{10} \right) \right) \\
 &= \frac{1}{12}
 \end{aligned}$$